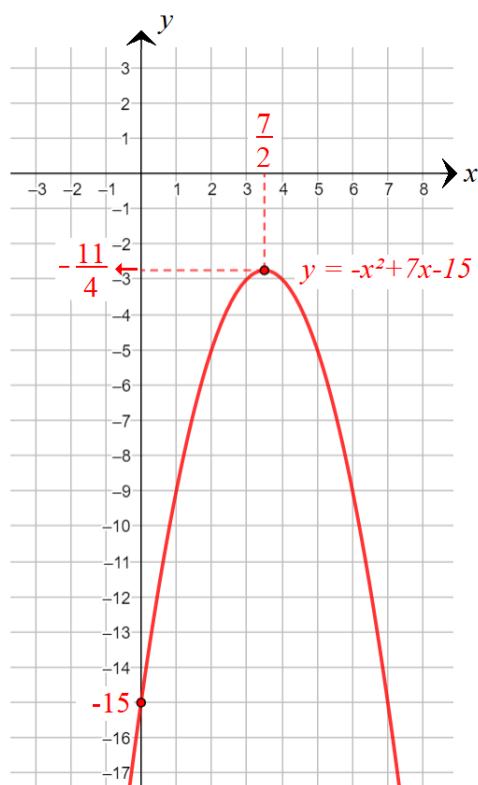


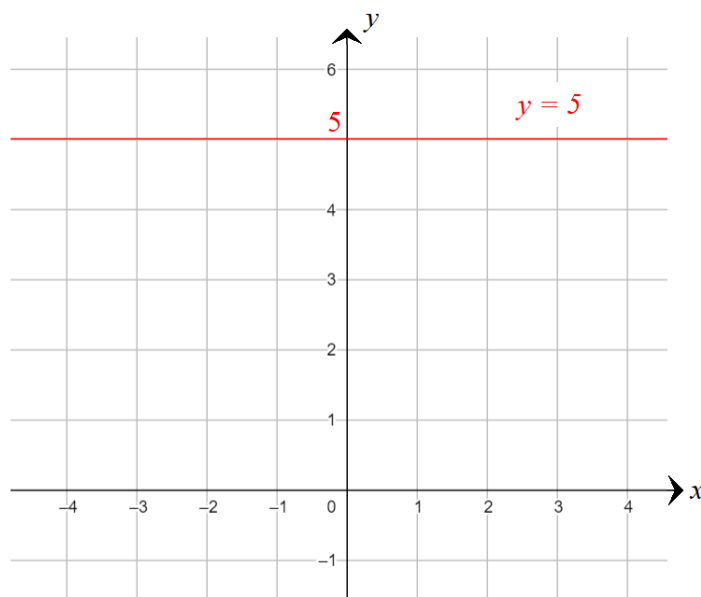
Gabarito da lista de exercícios 1

- (1) (a) $\left\{x \in \mathbb{R} \mid x \neq \frac{1}{7}\right\} = \mathbb{R} - \left\{\frac{1}{7}\right\} = \left(-\infty, \frac{1}{7}\right) \cup \left(\frac{1}{7}, +\infty\right).$
- (b) $\left\{x \in \mathbb{R} \mid x \geq -\frac{1}{5} \text{ e } x \neq \frac{4}{3}\right\} = \left[-\frac{1}{5}, +\infty\right) - \left\{\frac{4}{3}\right\} = \left[-\frac{1}{5}, \frac{4}{3}\right) \cup \left(\frac{4}{3}, +\infty\right).$
- (c) $\left\{x \in \mathbb{R} \mid x > \frac{9}{2}\right\} = \left(\frac{9}{2}, +\infty\right).$
- (d) $\left\{x \in \mathbb{R} \mid x \neq -\frac{3}{4}\right\} = \mathbb{R} - \left\{-\frac{3}{4}\right\} = \left(-\infty, -\frac{3}{4}\right) \cup \left(-\frac{3}{4}, +\infty\right).$
- (e) $\mathbb{R}.$
- (f) $\left\{x \in \mathbb{R} \mid -\frac{3}{2} \leq x \leq \frac{3}{2}\right\} = \left[-\frac{3}{2}, \frac{3}{2}\right].$
- (g) $\{x \in \mathbb{R} \mid x < -1 \text{ ou } x > 4\} = (-\infty, -1) \cup (4, +\infty).$
- (h) $\left\{x \in \mathbb{R} \mid -2 \leq x \leq 3 \text{ e } x \neq \frac{7}{5}\right\} = [-2, 3] - \left\{\frac{7}{5}\right\} = \left[-2, \frac{7}{5}\right) \cup \left(\frac{7}{5}, 3\right].$

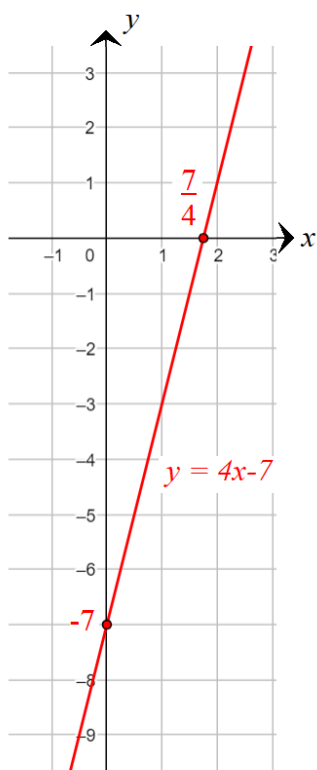
- (2) (a) $Dom(f) = \mathbb{R}$
 $Im(f) = \left(-\infty, -\frac{11}{4}\right]$



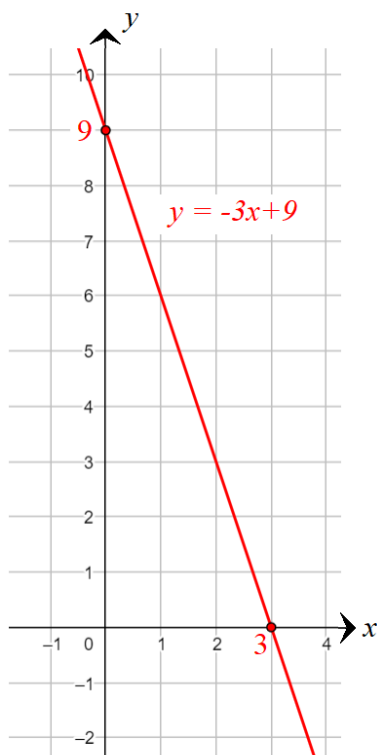
- (b) $Dom(f) = \mathbb{R}$
 $Im(f) = \{5\}$



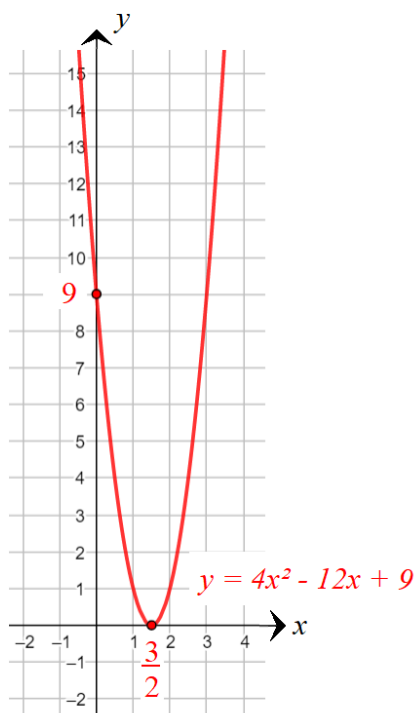
(c) $Dom(f) = \mathbb{R}$
 $Im(f) = \mathbb{R}$



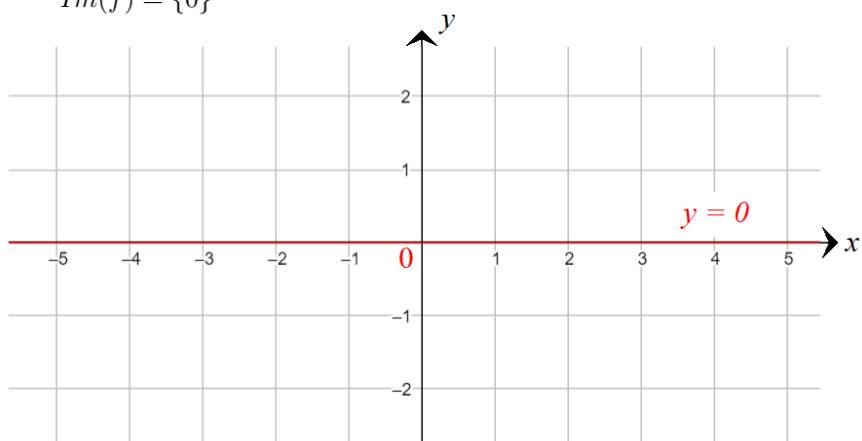
(d) $Dom(f) = \mathbb{R}$
 $Im(f) = \mathbb{R}$



(e) $Dom(f) = \mathbb{R}$
 $Im(f) = \left[\frac{3}{2}, +\infty\right)$

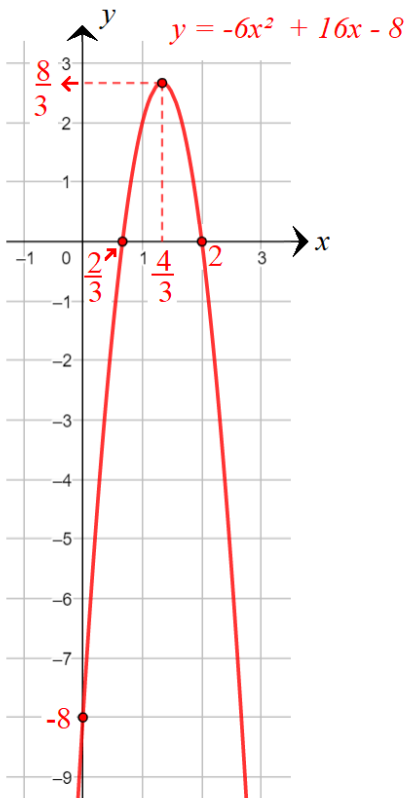


(f) $Dom(f) = \mathbb{R}$
 $Im(f) = \{0\}$



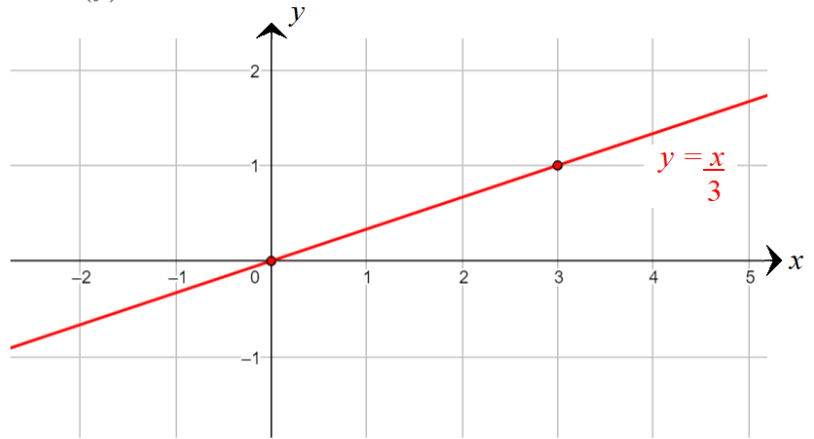
(g) $Dom(f) = \mathbb{R}$

$Im(f) = \left(-\infty, \frac{8}{3}\right]$



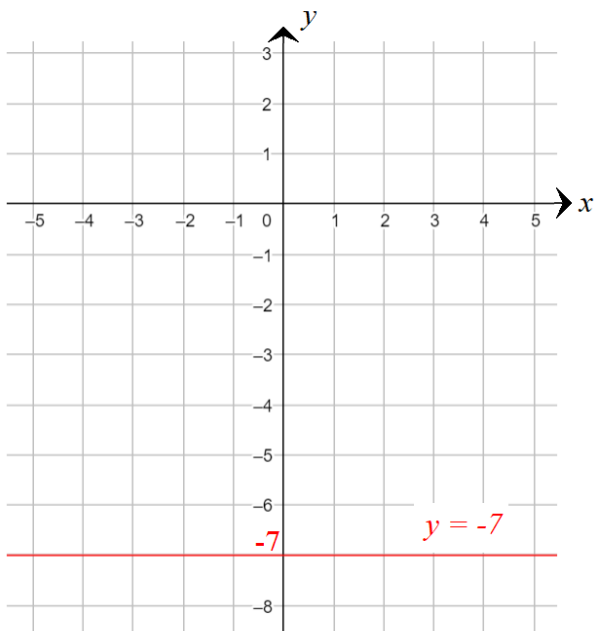
(h) $Dom(f) = \mathbb{R}$

$Im(f) = \mathbb{R}$



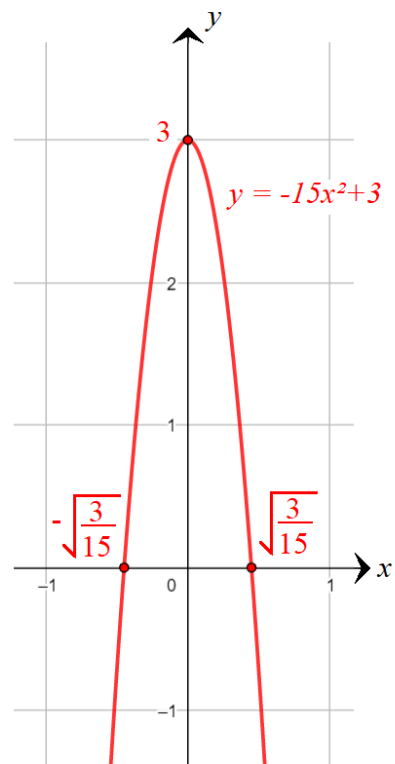
(i) $Dom(f) = \mathbb{R}$

$Im(f) = \{-7\}$

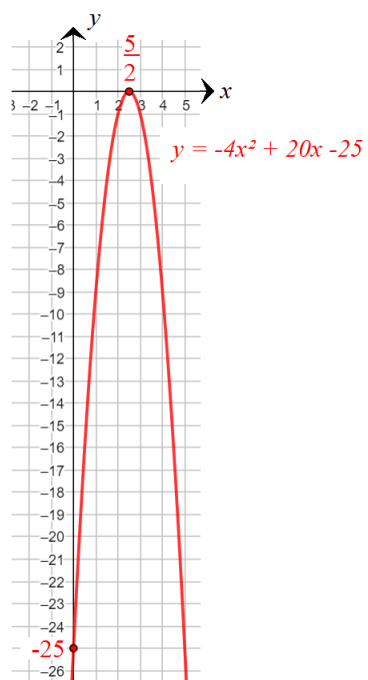


(j) $Dom(f) = \mathbb{R}$

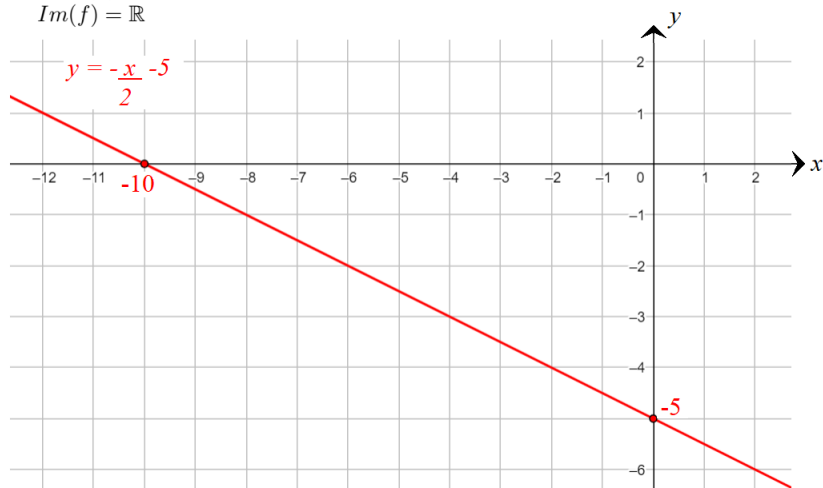
$Im(f) = (-\infty, 3]$



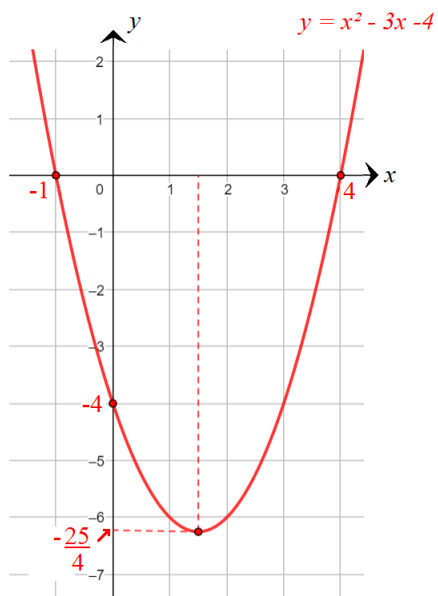
(k) $Dom(f) = \mathbb{R}$
 $Im(f) = (-\infty, 0]$



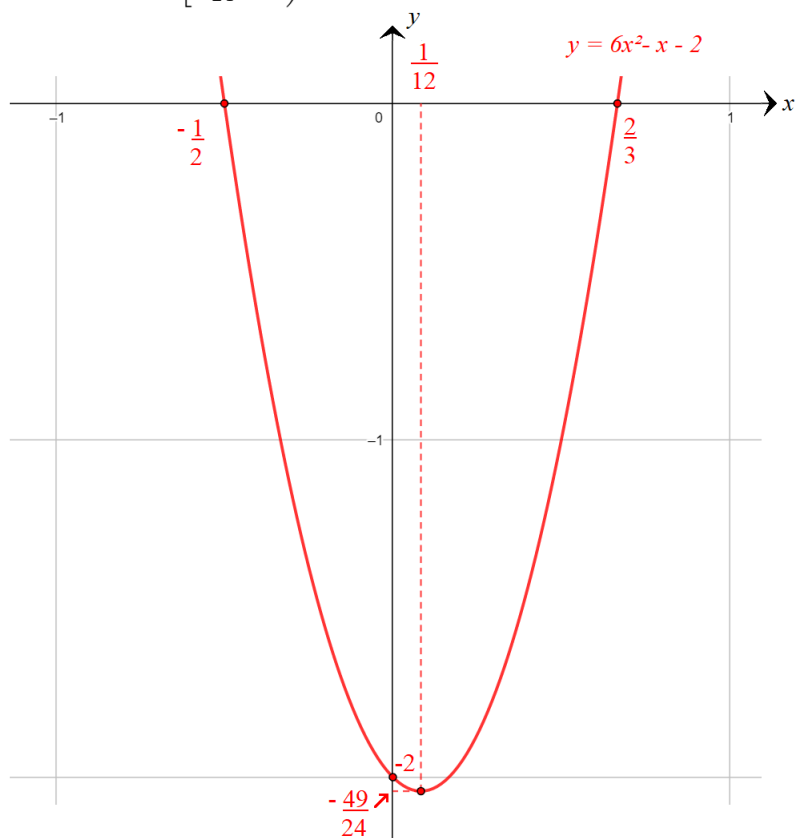
(l) $Dom(f) = \mathbb{R}$
 $Im(f) = \mathbb{R}$



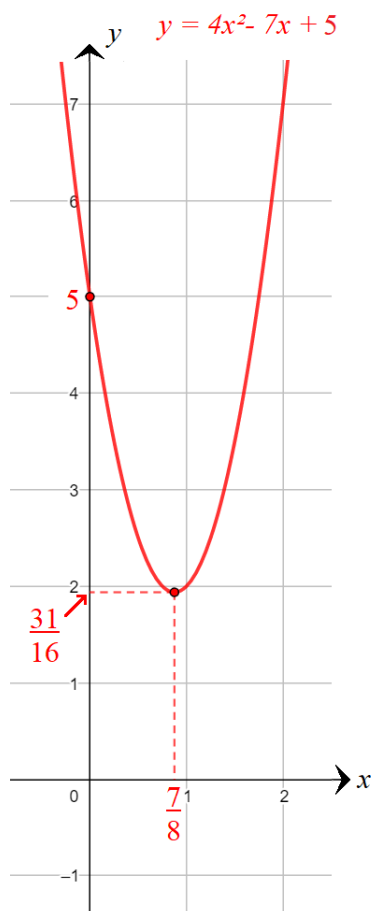
(m) $Dom(f) = \mathbb{R}$
 $Im(f) = \left[-\frac{25}{4}, +\infty\right)$



(n) $Dom(f) = \mathbb{R}$
 $Im(f) = \left[-\frac{49}{24}, +\infty\right)$



(o) $Dom(f) = \mathbb{R}$
 $Im(f) = \left[\frac{31}{16}, +\infty\right)$



(2) (a) $5^{-1} = \frac{1}{5}$.

(b) $3^{-50} = \frac{1}{3^{50}}$.

(c) 5.

(d) 0.

(3) $k = 4x + \frac{27}{2}$.

(4) $k = \frac{1400}{x}$.

(5) $k = \frac{21x}{4}$.

(6) $x = \frac{4}{3}$ e $x = -\frac{4}{3}$.

(7) $x = 3$.

(8) (a) $\mathbb{R}$.

(b) $(2, +\infty)$.

(c) $[3, +\infty)$.

(d) $\left\{x \in \mathbb{R} \mid x \neq \frac{\pi}{2}, n \in \mathbb{Z}\right\}$.

(e) $\left\{x \in \mathbb{R} \mid x \neq \frac{\pi}{2}, n \in \mathbb{Z} \text{ ímpar}\right\}$.

(f) $(-\infty, -\sqrt{10}] \cup [\sqrt{10}, +\infty)$.

(g) $((-\infty, -2) \cup (3, +\infty)) - \left\{x \in \mathbb{R} \mid x = \frac{n\pi}{2}, n \in \mathbb{Z} \text{ ímpar ou } x = \frac{1 - \sqrt{21}}{2} \text{ ou } x = \frac{1 + \sqrt{21}}{2}\right\}$.

(h) $[-1, 1] - \{0\}$.

(i) $\mathbb{R}^+$.

(9) (a) $\sin x = \frac{\sqrt{3}}{2}, \cos x = \frac{1}{2}, \tan x = \sqrt{3}, \operatorname{cosec} x = \frac{2}{\sqrt{3}}, \sec x = 2$.

(b) $\sin x = \frac{1}{2}, \cos x = \frac{\sqrt{3}}{2}, \tan x = \frac{1}{\sqrt{3}}, \operatorname{cosec} x = 2, \sec x = \frac{2}{\sqrt{3}}$.

(c) $\sin x = 1, \cos x = 0, \operatorname{cosec} x = 1, \frac{5\pi}{2} \notin \operatorname{Dom}(\tan x) \text{ e } \frac{5\pi}{2} \notin \operatorname{Dom}(\sec x)$.

(d) $\sin x = 0, \cos x = -1, \tan x = 0, \sec x = -1, 7\pi \notin \operatorname{Dom}(\operatorname{cosec} x)$.

(e) $\sin x = 0, \cos x = -1, \tan x = 0, \sec x = -1, -5\pi \notin \operatorname{Dom}(\operatorname{cosec} x)$.

(f) $\sin x = \frac{1}{2}, \cos x = -\frac{\sqrt{3}}{2}, \tan x = -\frac{1}{\sqrt{3}}, \operatorname{cosec} x = 2, \sec x = -\frac{2}{\sqrt{3}}$.

(g) $\sin x = -\frac{\sqrt{3}}{2}, \cos x = \frac{1}{2}, \tan x = -\sqrt{3}, \operatorname{cosec} x = -\frac{2}{\sqrt{3}}, \sec x = 2$.

(h) $\sin x = \frac{\sqrt{2}}{2}, \cos x = \frac{\sqrt{2}}{2}, \tan x = 1, \operatorname{cosec} x = \frac{2}{\sqrt{2}}, \sec x = \frac{2}{\sqrt{2}}$.

(i) $\sin x = \frac{\sqrt{2}}{2}, \cos x = -\frac{\sqrt{2}}{2}, \tan x = -1, \operatorname{cosec} x = \frac{2}{\sqrt{2}}, \sec x = -\frac{2}{\sqrt{2}}$.

(j) $\sin x = -\frac{\sqrt{3}}{2}, \cos x = \frac{1}{2}, \tan x = -\sqrt{3}, \operatorname{cosec} x = -\frac{2}{\sqrt{3}}, \sec x = 2$.

(k) $\sin x = -\sin a, \cos x = -\cos a, \tan x = \tan a, \operatorname{cosec} x = -\operatorname{cosec} a, \sec x = -\sec a$.

(l) $\sin x = \cos a, \cos x = -\sin a, \tan x = -\tan a, \operatorname{cosec} x = \sec a, \sec x = -\operatorname{cosec} a$.